

MONCTON ROMEO LEBLANC, NB, Canada

WMO: 717050

Lat: 46.1053N

Lon: 64.6839W

Elev: 232

StdP: 14.57

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.7	(c) -2.5	(d) -17.4	(e) 2.1	(f) -5.4	(g) -12.4	(h) 2.8	(i) -1.2	(j) 34.8	(k) 21.6	(l) 31.1	(m) 21.4	(n) 12.3	(o) 260	(p) 0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.9	(c) 83.9	(d) 69.7	(e) 80.9	(f) 67.8	(g) 78.2	(h) 66.3	(i) 72.1	(j) 80.3	(k) 70.5	(l) 77.5	(m) 68.8	(n) 75.1	(o) 13.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	109.2	76.0	67.9	103.5	74.4	66.3	97.8	72.6	36.0	80.2	34.6	77.7	33.1	75.2	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.6	(b) 24.8	(c) 21.9	(e) -12.6	(f) 89.7	(g) 5.3	(h) 2.6	(i) -16.4	(j) 91.6	(k) -19.6	(l) 93.1	(m) -22.5	(n) 94.6	(o) -26.4	(p) 96.5		
(a) 28.6	(b) 24.8	(c) 21.9	(e) -13.2	(f) 75.5	(g) 5.1	(h) 1.4	(i) -16.9	(j) 76.5	(k) -19.9	(l) 77.3	(m) -22.8	(n) 78.1	(o) -26.5	(p) 79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	17.9	19.6	27.7	39.0	50.4	59.8	67.0	65.9	58.3	47.0	36.3	25.2
	DBStd	18.86	11.59	10.48	8.87	7.24	6.99	6.62	4.76	5.11	6.96	7.37	8.95	9.97
	HDD50	4310	996	850	692	340	81	5	0	0	10	147	420	769
	HDD65	8265	1461	1270	1156	781	455	180	34	52	222	559	861	1234
	CDD50	1745	0	0	1	9	93	300	528	492	258	55	9	0
	CDD65	223	0	0	0	0	2	24	97	79	20	1	0	0
	CDH74	1759	0	0	1	3	59	250	735	573	135	3	0	0
	CDH80	348	0	0	0	0	7	45	151	118	27	0	0	0
Wind	WSAvg	10.5	12.0	11.7	12.0	11.3	10.2	9.3	8.5	8.4	9.2	10.3	11.3	11.8
Precipitation	PrecAvg	50.8	5.0	4.1	5.0	4.2	3.6	3.8	4.2	3.2	3.3	4.3	4.6	5.4
	PrecMax	63.5	11.2	6.5	7.9	9.4	8.8	6.8	8.5	7.1	6.2	8.6	9.1	10.4
	PrecMin	33.1	0.7	1.6	1.1	1.4	0.9	0.9	0.7	0.6	0.6	1.7	1.3	1.8
	PrecStd	7.5	2.3	1.1	1.7	2.0	1.9	1.5	2.0	1.5	1.4	1.8	1.7	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	50.1	56.1	68.3	80.4	85.0	87.6	87.5	83.5	73.1	63.9	55.1
		MCWB	52.9	47.4	47.7	56.6	61.9	68.8	72.5	71.5	70.3	63.2	60.4	53.3
	2%	DB	45.5	42.9	48.3	60.9	74.1	80.1	84.0	83.2	77.5	66.8	59.4	48.5
		MCWB	44.0	40.7	43.5	52.5	58.9	66.0	70.6	69.2	66.9	59.4	56.7	47.3
	5%	DB	39.2	38.3	44.1	56.0	68.9	76.7	81.1	80.0	73.7	63.0	55.4	43.0
		MCWB	37.1	35.6	40.0	49.1	56.6	64.3	68.6	67.6	65.1	58.1	51.9	41.3
	10%	DB	33.9	34.6	40.1	51.4	64.0	72.9	78.2	76.9	70.2	59.4	49.9	38.5
		MCWB	32.5	32.8	36.3	44.8	54.4	62.0	66.9	66.2	63.3	54.5	46.5	36.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	48.0	50.8	59.8	65.8	72.0	74.6	74.2	72.0	65.3	61.7	53.2
		MCDB	54.4	49.9	53.3	64.9	75.0	81.1	84.7	82.4	80.5	69.0	63.6	54.4
	2%	WB	44.3	41.2	45.1	54.8	61.8	69.1	72.4	71.7	69.3	62.0	57.3	47.6
		MCDB	45.2	42.7	47.6	58.6	69.6	76.5	81.1	79.1	74.6	65.0	58.8	48.7
	5%	WB	37.6	36.1	40.6	50.1	58.8	66.5	70.7	70.0	67.1	58.8	52.6	41.7
		MCDB	38.9	37.9	43.2	54.6	65.9	72.8	78.1	76.9	71.7	62.6	54.9	42.8
	10%	WB	32.6	32.9	36.6	45.7	55.6	64.0	68.9	68.1	64.6	55.2	47.2	37.0
		MCDB	33.7	34.3	39.7	50.8	62.1	70.4	75.4	74.4	69.1	58.3	49.4	38.2
Mean Daily Temperature Range	5% DB	MDBR	16.2	16.3	15.2	16.2	19.4	19.3	18.9	19.1	18.9	16.5	13.5	13.7
		MCDBR	23.7	18.6	19.7	24.5	28.9	25.9	23.8	23.8	22.6	21.7	18.9	18.1
		MCWBR	23.6	17.0	16.1	16.9	16.3	14.4	12.3	12.0	13.4	15.6	17.3	17.8
	5% WB	MCDBR	23.6	18.0	18.7	21.3	24.0	21.4	20.3	19.7	19.3	18.7	18.1	18.1
		MCWBR	23.8	17.1	17.0	17.5	15.8	13.5	11.9	11.6	12.9	15.4	17.7	18.2
Clear-Sky Solar Irradiance	taub	0.287	0.284	0.319	0.344	0.366	0.387	0.414	0.388	0.354	0.326	0.306	0.284	
	taud	2.413	2.399	2.402	2.419	2.403	2.343	2.302	2.370	2.452	2.530	2.530	2.496	
	Ebn at Noon	260	286	289	290	285	279	269	272	273	267	252	250	
	Edh at Noon	22	27	32	34	36	39	40	36	30	24	20	19	
All-Sky Solar Radiation	RadAvg	455	694	1072	1344	1577	1703	1750	1543	1231	778	433	335	
	RadStd	41	62	69	131	165	135	119	104	102	78	50	30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (42 neighbors)	N/A	N/A	N/A	N/A	+0.68	+0.85	N/A	N/A	+117	+43
	N/A	N/A	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page